Launcher Test

Does the flywheel spin the right way, where should the trigger gate sit, when is the wheel READY, and do shots land? (issue #5, decisions 029, 032)

Names		Date	
Battery voltage at start		Battery at end	

OpMode: Launcher Test (Tests group on the Driver Station). **Needs in the config:** `launcherMotor` (Motor, exact goBILDA type), `launcherTriggerServo` (Servo)

Hidden by default (052): delete `@Disabled` in its file, build, deploy. **Safety:** space around the robot, hands clear, one person on STOP. Nothing moves in INIT.

The flywheel is fast and the ball comes out hard. Nobody in front of the launcher. Aim at a wall or a net for steps 1 to 5. Eye protection on.

Controls (gamepad 1)

Button	Does
A	start / stop the flywheel at the test power
B	STOP the flywheel
Dpad UP / DOWN	test power +10% / -10%
X / Y	test power 50% / 100%
HOLD right trigger	power follows the trigger (0 to 100%)
Dpad LEFT / RIGHT	flip the motor direction (for step 1)
HOLD left bumper	trigger gate to FIRE (open). Let go = REST (closed)

Steps, in order

#	Do this	Expected	Actual (what happened)	If wrong
1	X, then A	wheel spins so a ball would go OUT the launcher		flip with dpad; if FORWARD is right, `MOTOR_REVERSED = false`
2	Wheel stopped. Put a ball in the path, let go of the left bumper	gate at REST BLOCKS the ball		change `TRIGGER_REST_POSITION`
3	Hold the left bumper	gate at FIRE lets the ball pass freely		change `TRIGGER_FIRE_POSITION`
4	Y, then A. Wait 3 s. Read "Launcher Velocity"	a steady number: _____ ticks/s		`READY_VELOCITY_THRESHOLD` = about 90% of it: _____
5	B, then A again. Count seconds until "READY"	under 2 s		note it; the operator must wait this long

#	Do this	Expected	Actual (what happened)	If wrong
6	Launch one ball (A, wait READY, hold left bumper)	ball comes out		jam: check the gate positions and the water wheel

Power vs distance

Put the robot a set distance from the goal. For each power, launch 5 balls. The hit-rate test at the end uses the best power.

Power	Distance to goal (in)	Made / 5	Too short / too long?	Notes
60%				
80%				
100%				
___%				

Hit-rate test (the "done when" of issue #5)

Power	Distance (in)	Battery (V)	Made / 10	Who launched

Numbers to put into the code: `subsystems/Launcher.java`

Constant	Was	New value	Checked by
`MOTOR_REVERSED`	true		
`TRIGGER_REST_POSITION`	0.0		
`TRIGGER_FIRE_POSITION`	1.0		
`READY_VELOCITY_THRESHOLD`	1000.0		
`DEFAULT_SPIN_POWER`	1.0		

Also write in `hardware-config.md`: motor gear ratio _____, Expansion Hub motor port ____, trigger servo hub ____ port ____.

After the test: a coder puts the new numbers into the constants and removes the matching TODO comments; the results go on GitHub issue #5. Photos of anything surprising!